

TWO-NESS CLINICS: The Receipt Test

Core question: Can you tell them apart later? Only if you kept the receipt.

Time: 7 min | **Audience:** Museum/Public | **Tier:** 0

Materials

- Two distinct tokens (e.g., beads or marked coins: **A** and **B**)
- Opaque capsule or small box
- Sealed **Audit Envelope** (a card that records a mapping, e.g., “Token A → Left; Token B → Right”)
- Optional: tablet or sticky-notes for a quick tally

Script & flow (tick pass/fail boxes during the demo)

0) Hook (30 sec)

“How many things are here? Let’s find out—and watch how the answer changes when we change the rules for keeping track.”

Show: two labeled tokens (A & B). **Ask:** “How many things?” **Expected:** “Two.”

Say: “Great. That label is the **receipt**: it tells A from B.”

1) Remove the receipt (60 sec)

Do: hide/remove labels so the tokens look identical.

Ask: “Now how many?” **Expected spread:** some “two,” some “one type (two tokens),” some hesitation.

Say: “Without the receipt, we lose *which-is-which*. You may still see two objects, but token-identity is gone.”

☐ **PASS:** Audience count-judgment shifts (toward “one type / multiplicity”).

2) Aggregate (60 sec)

Do: place both tokens into the opaque capsule, shake, and hold it up.

Ask: “Now?” **Expected:** “one object / one lump.”

Say: “Now even multiplicity is hidden. Many past states map to the same present description: two-ness has **operationally** disappeared.”

☐ **PASS:** Majority reports “one object.”

3) The audit toggle (90 sec) — the key

Do: hold up the sealed Audit Envelope.

“But what if I kept a receipt? This envelope recorded which token went where.”

Ask: “Does this change what *is*, or just what we can *know*?” (brief discussion)

Punchline: “With the receipt, the map from past → present is **invertible**. Without it, the ‘erasure’ is final.”

☐ **CRITICAL PASS:** With audit: reconstruction judged **POSSIBLE**.

☐ Without audit: reconstruction judged **IMPOSSIBLE**.

4) Link to physics (60 sec)

“This isn’t just philosophy. If you physically **erase** a physical receipt (reset a memory), there’s a minimum energy cost—often summarized as **kT ln 2** per bit. Keeping a side-channel avoids paying that cost *until you erase it*.”

Optional visual: a 10-second loop of the HOM dip disappearing/reappearing as which-path marks are added/removed.

5) Handoff (30 sec)

“Want to run the experiment yourself and contribute to our live data? Grab a worksheet, or scan the QR code to see the dashboard update in real time.”

TWO-NESS CLINICS: Data Sheet

Can you tell them apart later? Only if you kept the receipt.

Name (optional): _____

Date: _____

Station: D-_____

Part 1: Count judgments — what basis are you using?

Instructions: For each scenario, circle **one** primary response and rate your confidence (0 = guessing, 10 = absolutely certain).

Scenario A — Label basis (the receipt is obvious)

Two clearly different, labeled tokens (A & B).

Number of THINGS: [2] [1 type (2 tokens)] [1 object] Other: _____

Confidence (0-10): _____

Scenario B — Type basis (the receipt is removed)

The labels are hidden. The tokens look identical.

Number of THINGS: [2] [1 type (2 tokens)] [1 object] Other: _____

Confidence (0-10): _____

Scenario C — Aggregate basis (the tokens are mixed)

Both tokens are placed into a single opaque capsule.

Number of THINGS: [2] [1 type (2 tokens)] [1 object] Other: _____

Confidence (0-10): _____

Part 2: The audit toggle — does the receipt change reversibility?

After aggregating (Scenario C), consider two cases:

Case 1: Audit exists

A sealed record is kept of “Token A → Left side, Token B → Right side.”

Could you, in principle, reconstruct which was which? [YES] [NO]

Why or why not? (one sentence)

Case 2: No audit

No record was kept.

Could you, in principle, reconstruct which was which? [YES] [NO]

Why or why not? (one sentence)

TWO-NESS CLINICS: Data Sheet (continued)

Part 3: Prediction check

Circle your final judgment.

- 1) If the audit is **kept**, the “loss” of two-ness is: [REAL] or [APPARENT]
 - 2) If the audit is **erased**, the “loss” of two-ness is: [REVERSIBLE] or [IRREVERSIBLE] (operationally)
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Part 4 (optional): Physics connection — HOM effect

If you visited the quantum optics station, record the measured visibility (V).

V (unmarked, initial): _____

V (with which-path mark): _____

V (mark removed, final): _____

PASS/FAIL: Visibility drops when marked and returns when unmarked. [PASS] [FAIL]

Thank you! Your responses help test how humans individuate objects.

Scan the QR code to see the collective results update in real time.

QR CODE
placeholder

Dashboard link: <https://tinyurl.com/twonese-data-dashboard>